

# Ananth Shreekumar

Ph.D. Student, Department of Computer Science  
Purdue University, West Lafayette, IN

<https://ashreeku.github.io>  
[ashreeku@purdue.edu](mailto:ashreeku@purdue.edu)  
+1 (765) 767 1346

## EDUCATION

---

### Doctor of Philosophy in Computer Science

*Purdue University*

- Co-Advisors: Professor Z. Berkay Celik, Professor Dongyan Xu

**Present**

*West Lafayette, IN*

### Master of Science in Computer Science

*Purdue University*

- GPA : 3.93 / 4.0

**Dec 2023**

*West Lafayette, IN*

### Integrated Master of Technology in Computer Science and Engineering

*International Institute of Information Technology Bangalore (IIIT-B)*

- GPA : 3.92 / 4.0

**Jul 2021**

*Bangalore, India*

## EXPERIENCE

---

### American Express

*Software Engineer, Enterprise Architecture Team*

- Integrated machine learning to internal logging and monitoring framework to detect anomalies from logs.
- Implemented a pipeline to perform automated log analysis on real-time log data.

**Aug 2021 - Dec 2021**

*Bangalore, India*

### Siemens Healthineers

*Technical Intern, Center for Innovation in Diagnostics Team*

- Designed a deep learning pipeline to predict Sepsis onset in ICU patients from bedside measurements.
- Model performance exceeded 87% on accuracy, F1-score, specificity, and sensitivity metrics.

**Jan 2021 - Jul 2021**

*Bangalore, India*

## PUBLICATIONS

---

### Conference Publications

- C3 Ananth Shreekumar, JyunJhu Syu, Muslum Ozgur Ozmen, Dongyan Xu, Z. Berkay Celik. A Self-Explainable Deep Architecture for Security Applications. *Conference on Game Theory and AI for Security*. 2026.
- C2 Devin Ersoy\*, Brandon Lee\*, Ananth Shreekumar, Arjun Arunasalam, Muhammad Ibrahim, Antonio Bianchi, and Z. Berkay Celik. Investigating the Impact of Dark Patterns on LLM-Based Web Agents. *IEEE Symposium on Security and Privacy*. 2026.
- C1 Ananth Shreekumar\*, Biswesh Mohapatra\*, and Shrisha Rao. Incorporating Autonomous Bargaining Capabilities into E-Commerce Systems. *ACM International Conference on Intelligent Virtual Agents*. 2020.

### Journal Articles

- J1 Ankit Gupta, Ruchi Chauhan, Saravanan G., and Ananth Shreekumar. Improving Sepsis Prediction in Intensive Care with SepsisAI: A clinical decision support system with a focus on minimizing false alarms. *PLOS Digital Health*. 2024.

### Workshop Publications

- W2 Edoardo Allegrini, Ananth Shreekumar, and Z. Berkay Celik. Formalizing the Safety, Security, and Functional Properties of Agentic AI Systems. *Agents in the Wild: Safety, Security, and Beyond Workshop, International Conference on Learning Representations*. 2026.
- W1 Tarun Dutt, G.N. Srinivasa Prasanna, Tathagato Rai Dastidar, and Ananth Shreekumar. Towards Artifact Rejection in Microscopic Urinalysis. *Medical Imaging Meets NeurIPS Workshop, Neural Information Processing Systems*. 2019.

## Preprints

- P3 Doguhan Yeke\*, Elif Su Temirel\*, Ananth Shreekumar\*, Brandon Lee, Dongyan Xu, Z. Berkay Celik. The Yes-Man Syndrome: Benchmarking Abstention in Embodied Robotic Agents. *arXiv preprint arXiv:2605.20544*. 2026.
- P2 Zengxiang Lei, Ananth Shreekumar, Jonathan Rosenthal, Ruoyu Song, Alvaro A. Cardenas, Daniel J. Fremont, Dongyan Xu, Satish Ukkusuri, Z. Berkay Celik. Stable GFlowNets with TV Monitoring and Probabilistic Guarantees. *arXiv preprint arXiv:2605.01729*. 2026.
- P1 Mohammad Atif Quamar, Mohammad Areeb, Nishant Sharma, Ananth Shreekumar, Jonathan Rosenthal, Muslum Ozgur Ozmen, Mikhail Kuznetsov, Z. Berkay Celik. Adaptive Blockwise Search: Inference-Time Alignment for Large Language Models. *arXiv preprint arXiv:2510.23334*. 2025.

\* Equal contribution.

## SERVICE

---

### External Reviewer

SIGKDD (2026, 2027), NDSS (2025, 2026), IEEE S&P (2025, 2026), CCS (2025), NeurIPS (2025), AAAI (2026), ICLR (2026)

### Outreach

- Presented research to about 20 high school students as part of a 4-H event. 2024, 2025.

### Graduate Teaching Assistant

- |                                            |             |
|--------------------------------------------|-------------|
| • CS361 Great Issues in Computer Science   | Spring 2025 |
| • CS529 Security Analytics                 | Fall 2024   |
| • CS240 Programming in C                   | Fall 2023   |
| • CS352 Compilers: Principles and Practice | Spring 2023 |
| • MA165 Analytic Geometry and Calculus I   | Fall 2022   |

### Guest Lecturer

- |                                          |                      |
|------------------------------------------|----------------------|
| • CS197 Freshman Honors Seminar          | Spring 2026          |
| • CS529 Security Analytics               | Fall 2025, Fall 2024 |
| • CS361 Great Issues in Computer Science | Spring 2025          |

## REFERENCES

---

### Z. Berkay Celik

Associate Professor of Computer Science  
Purdue University  
Email: [zcelik@purdue.edu](mailto:zcelik@purdue.edu)  
Phone: +1 (765) 496-1761

### Dongyan Xu

Samuel D. Conte Professor of Computer Science  
Purdue University  
Email: [dxu@purdue.edu](mailto:dxu@purdue.edu)  
Phone: +1 (765) 494-7841